EE23 Term Project Proposal: Design of Audio-to-MIDI Linear Processing System with Matlab DSP

By Jacob Gerson, Charlotte Lokere, Mason Sibley

Introduction

Our project aims to leverage Matlab's internal DSP functions to describe audio signals as time-varying functions that can be manipulated and processed through LTI systems. In this project we are aiming to take advantage of Matlab's DSP toolbox to create a filter that extracts specific frequency components from an MP3 input and converts them into discrete MIDI signals.

Traditional MIDI interfaces rely on discrete mechanical/digital triggers from instruments - our model will instead respond to raw audio signals in the continuous time domain. By leveraging Matlab as a frequency-domain analysis tool, we'll be extracting specific harmonic components from audio inputs, like an MP3 file, and mapping them to discrete MIDI signals.

Linear System Modeling

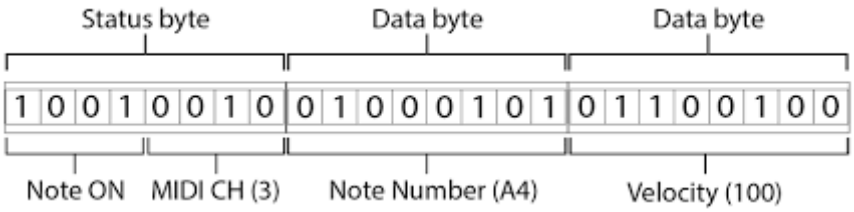
We'll model the audio analysis stage as a linear time invariant system which operates on discrete time audio signals derived from an MP3 input. Our pipeline will convert raw MP3 audio into discrete MIDI signals in a few stages. First the audio will be decoded into normalized discrete-time signals and then segmented into overlapping, windowed frames. Next a series of LTI digital filters will be used to pull out the strength of each frequency band over time tied to MIDI pitches, and this will feed into a decision layer which will generate note on/off events, velocities, and timings. This can then finally be compiled into a MIDI file for playback or further analysis, depending on the final application.

Implementation

Our implementation plan is described below. Essentially, a stream of audio signals (presumably MP3 data) would first be pre-processed in Matlab. This pre-processing would include signal normalization, bandpass filtering, noise reduction, and several other acoustic components. This normalized signal would then pass through the designed DSP filter system, which will include algorithms that recognize and

capture key MIDI data (note on, pitch, note velocity, pitch bend). This DSP filter will be the LTI system of importance in this project, and existing Matlab DSP functions will need to be tweaked and amended to fit our signal needs.

These note events (note on and note off are equally important in MIDI land!) will then be exported as the output of our software model of a linear system!



An example MIDI Data String

Proposed Go Beyond

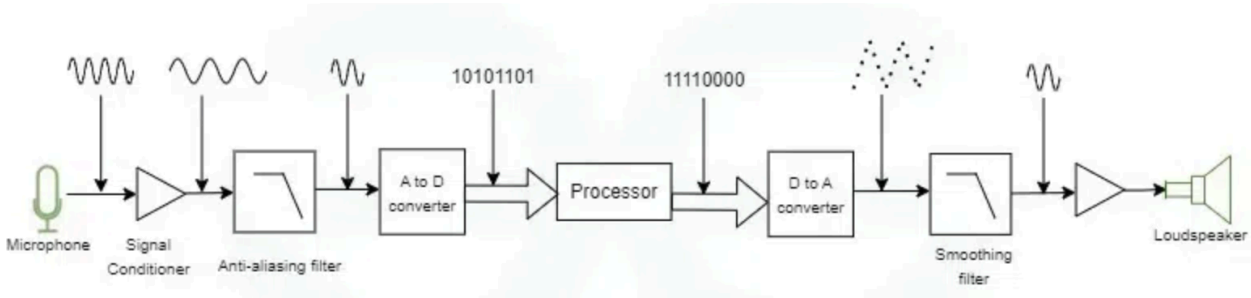
Though the above described software system covers the bases for this project proposal, we are very interested in bringing this sort of project up to something much more tangible.

Teammate Mason Sibley has already built a MIDI processor for his “Intro to Digital Logic” class this school year - it would be amazing and very much intuitive to connect these 2 projects!

This would mean including 2 new, and fairly easy, steps into our implementation strategy. From the generated MIDI data in Matlab, all we’d plan to do would be streaming that data through to Mason’s MIDI processor, which already does much of the binary encoding to play the requisite audio through a connected speaker.

The proposed final design:

Audio Signals -> DSP System -> MIDI -> MIDI Processor -> Speaker



Our group is incredibly excited about this project, and we’d be all the more proud to eventually see the linear system we design end up playing MIDI audio in real-time.